

Research Statement

Yichao Jin, Ph.D. | yichao.jin@utdallas.edu | yichao2022.github.io

Methods: Statistical Modeling, Simulation, AI/ML, Causal Inference, Bayesian Methods

Quantitative Modeling and Simulation

My research centers on building and validating quantitative models of complex decision processes using computational simulation, statistical inference, and machine learning. I specialize in translating empirical data into structural models that can generate counterfactual predictions — a skill set directly applicable to regulatory science and drug evaluation.

My dissertation developed the **Behavioral Data Toolkit (BDT)**, a three-tier computational framework that integrates experimental choice data with structural simulation. The framework involves Bayesian parameter estimation, Monte Carlo simulation for counterfactual generation, and AI/ML-based validation of model outputs. While the application domain was health decision-making, the methodological core — structural modeling, simulation, statistical validation, and AI integration — is domain-agnostic.

AI/ML and Data Science Skills

DQMM's research priorities include integrating AI/ML models with conventional pharmacokinetic modeling. My direct experience includes:

- **Machine learning for behavioral prediction:** Applied LLM-based synthetic agents to simulate human decision patterns, achieving 93.8% reduction in prediction error through empirical-frontier regularization.
- **Bayesian inference and structural modeling:** Built hierarchical Bayesian choice models incorporating random parameters, with posterior predictive checks for model validation.
- **Causal inference with observational data:** Experience with difference-in-differences, instrumental variables, and matching methods using large-scale survey and administrative datasets.
- **Computational tools:** Proficient in R (tidyverse, brms, Stan), Python (PyTorch, scikit-learn, pandas), and reproducible research workflows (Git, Quarto, LaTeX).

Connection to Regulatory Science

The BDT framework addresses a problem that parallels challenges in generic drug evaluation: how to use limited empirical data to make reliable predictions about unobserved scenarios. In drug development, this is the question of whether *in silico* modeling can reduce or replace *in vivo* bioequivalence studies; in my work, it is the question of whether computational simulations can pre-screen policy interventions before field implementation. The quantitative toolkit — structural model specification, sensitivity analysis, uncertainty quantification, and external validation — is shared.

I recognize that pharmacometric modeling (PK/PD, PBPK) requires domain knowledge I do not yet possess. What I offer is strong foundations in the *methodology* of quantitative modeling and simulation: model building, statistical inference, computational implementation, and rigorous validation. I am eager to apply these skills to DQMM's regulatory science mission and to rapidly acquire the domain-specific knowledge needed.

Proposed Contribution

I would contribute to DQMM's research by bringing expertise in statistical modeling, simulation methodology, and AI/ML integration — supporting the Division's work on quantitative approaches to generic drug evaluation while developing regulatory science skills under Dr. Babiskin's mentorship.

Selected Publications: Jin, Y. (2026). *Empirical-Frontier Regularization for LLM Synthetic Agents*. SSRN. | Jin, Y. (2026). *Waiting Time as a Behavioral Barrier to Vaccination Uptake*. Job Market Paper.